

Literature Status: the CAR Foguel–Hankel B_2 Condition

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Status. `literature_already_answered`.

Original question. For a scalar sequence $\alpha = (\alpha_k)_{k \geq 0}$, Badea’s *Operators near completely polynomially dominated ones and similarity problems* [1] considers the associated CAR-valued Foguel–Hankel operator $R(Y_\alpha)$ and sets

$$B_2(\alpha) = \sum_{k \geq 0} (k+1)^2 |\alpha_k|^2.$$

Theorem 1.4 proves that similarity of $R(Y_\alpha)$ to a contraction implies $B_2(\alpha) < \infty$. Immediately afterward, on page 5 of the 22-page arXiv PDF, the source asks:

It is an open problem if B_2 finite implies $R(Y_\alpha)$ similar to a contraction.

The question is repeated after Corollary 5.6 on page 14.

Identification of the answer. Ricard’s distinct paper *On a Question of Davidson and Paulsen* [2] supplies the affirmative converse. Its abstract explicitly identifies the result as an answer to the Davidson–Paulsen question about similarity to a contraction for this class of CAR-valued Foguel–Hankel operators. Thus, together with the already-known necessity,

$$R(Y_\alpha) \text{ is similar to a contraction} \iff \sum_{k \geq 0} (k+1)^2 |\alpha_k|^2 < \infty.$$

A later open-access source gives independent full-text corroboration: Chalmoukis–Stylogiannis [3], Theorem 4.1 and the paragraph immediately following it (PDF pages 14–15), state Ricard’s Hankel Schur-multiplier theorem and explicitly say that this theorem answers the Davidson–Paulsen question about CAR-valued Foguel–Hankel operators similar to a contraction.

Verification note. Ricard’s HAL record contains the DOI and journal metadata but currently no deposited file, and the publisher full-text API requires a private key. This packet therefore does not claim direct local inspection of the journal PDF; it relies on the answer-identifying journal abstract plus the directly inspected later full-text attribution. This is an explicit later-literature answer, not a new theorem of this run.

References

- [1] C. Badea, *Operators near completely polynomially dominated ones and similarity problems*, Journal of Operator Theory **49** (2003), 3–23; arXiv:math/0102160.

- [2] E. Ricard, *On a Question of Davidson and Paulsen*, Journal of Functional Analysis **192** (2002), 283–294; doi:10.1006/jfan.2001.3899.
- [3] N. Chalmoukis and G. Stylogiannis, *Quasi-nilpotency of Generalized Volterra Operators on Sequence Spaces*, Results in Mathematics **76** (2021), 173; arXiv:2005.01660.